

Trainings ▾

General Information

The QSK45 and QSK60 series engines use a positive displacement, rotary screw-style lubricating oil pump. The lubricating oil pump incorporates a high-pressure relief valve, regulator valve, pumping screws, and drive train all in one assembly. The lubricating oil pump housing is precision bored to accept the power rotor screw and idler rotor screws. When rotated, the rotors form cavities and fill with oil; the oil is moved axially from the inlet to the outlet. As the oil is moved from the inlet to the outlet the cavities start to close, putting the oil under pressure.

The lubricating oil pump is mounted at the front of the cylinder block and driven via an idler gear by the crankshaft gear. The ratio of the gearing is 1.2:1. The lubricating oil pump rotates **clockwise**, the same as the crankshaft.

Proper alignment and gear backlash are critical to the durability of the lubricating oil pump. Backlash is adjusted using shims between the feet of the pump and the cylinder block mounting surface. **Always** check the gear backlash between the idler gear and the crankshaft gear with the engine in the normal upright position. Lubricating oil pump failure can occur if the backlash is **not** set correctly.

Note : The lubricating oil pump for the QSK45 and QSK60 engines look the same, but they are **not** interchangeable.

Lubricating Oil Suction Tube

Several different suction tube options are available to accommodate the various sump depths and locations available for the engines. The suction tube **must** match the oil pan sump option for proper lubrication.

No gaskets are used between the lubricating oil pump and block connections.

The **only** gasket used is on the lubricating oil pump inlet housing where the suction tube connects to the pump. The housing is retained by socket head capscrews that use a thread lock compound for retention of the capscrews. If this area is **not** sealed properly, a low oil pressure situation can occur and cause engine damage.

The lubricating oil transfer tube is mounted between the oil pump and the engine block connections. Square cut o-rings and three bolt clamps are used to seal the tube into the connectors. Orientation of the tube is required during assembly for proper alignment.

The QSK45 and QSK60 engines have larger oil pan adapters. Engines can use either the “chart method” or the standard service intervals for the oil change intervals. The chart method is totally based on load factor (fuel usage), lubricating oil quality, and lubricating oil system capacity. The QSK45 and the QSK60 engines often have one sump added to the pan.

Lubricating Oil Filter

The QSK45 and QSK60 engines use spin-on oil filters mounted on the right side of the engine. The QSK45 uses three filters and the QSK60 uses four filters of Venturi filter design, which provide full-flow and bypass lubricating oil filtration.

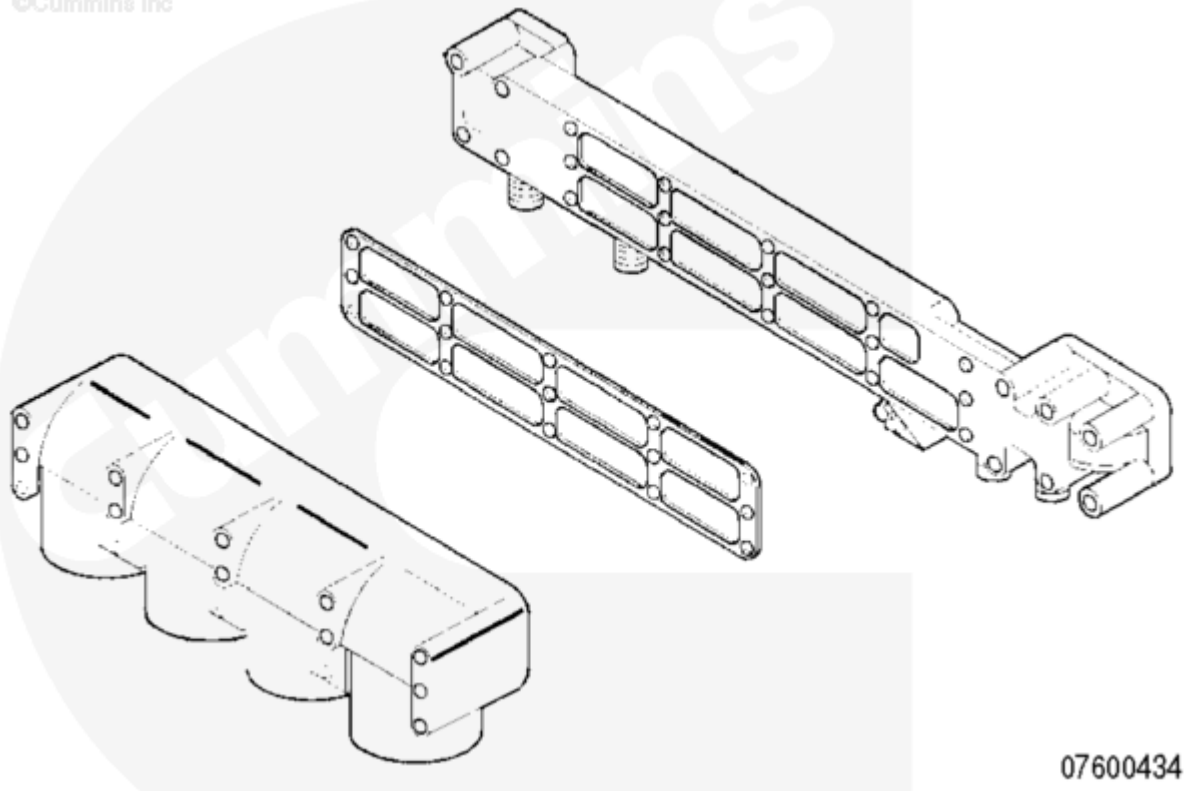
The Fleetguard® ConeStac™ centrifuge bypass filter is offered with the Venturi filters as standard equipment. Oil life and filter life are improved because of the high efficiency of the centrifuge.

The Eliminator filter is a full-flow, bypass combination oil filter. This unit bolts directly onto the engine block in place of the standard lubricating oil filter head. Both the QSK45 and QSK60 engines can use this full-flow bypass filter mounted on the right-bank side of the engine. Refer to Service Bulletin 3666253.

The Centinel™ is a duty-cycle-dependent lubrication management system that burns used engine oil with the fuel. New oil is added to the sump as the used engine oil is burned. The replacement of used oil with new oil stabilizes the oil quality and can significantly increase the oil change intervals and filter life. Refer to the Master Repair Manual - Centinel™, Bulletin 3666231.

One-Piece Filter Head, Spin-on Filter

The previous filter head design was a two piece assembly consisting of a back plate and a filter head. The back plate attaches to the cylinder block and the filter head attaches to the back plate. The previous two-piece filter head design is illustrated on the next page.



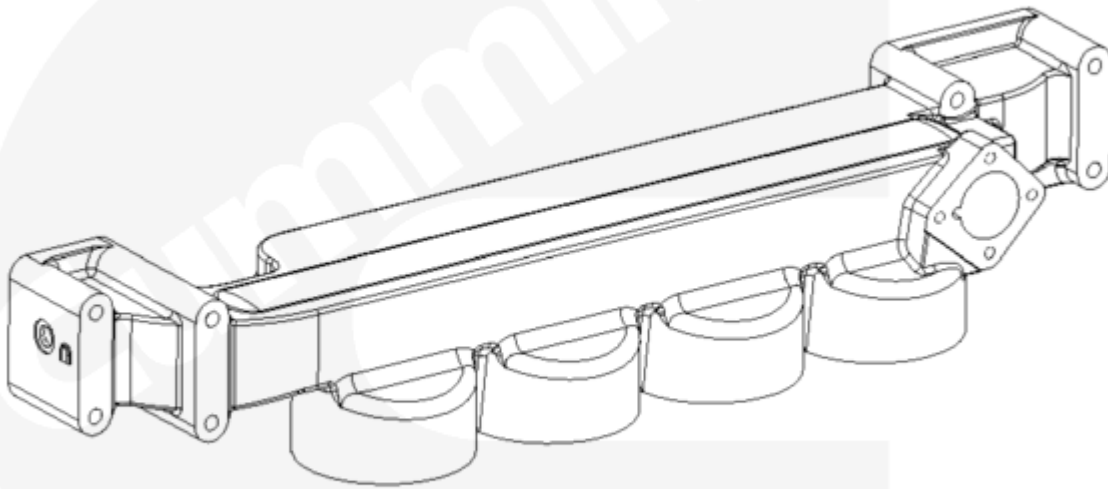
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Previous Two-Piece Filter Head Design

The updated filter head is a one-piece design that attaches directly to the cylinder block eliminating the joint between the filter head and back plate. The QSK45 filter head mounts three oil filters and the QSK60 filter head mounts four oil filters. The previous two-piece filter head has been obsoleted and replaced with the new one-piece filter head.

If the two-piece filter head number is ordered, the one-piece filter head will be received.

The updated one-piece filter head is illustrated below.



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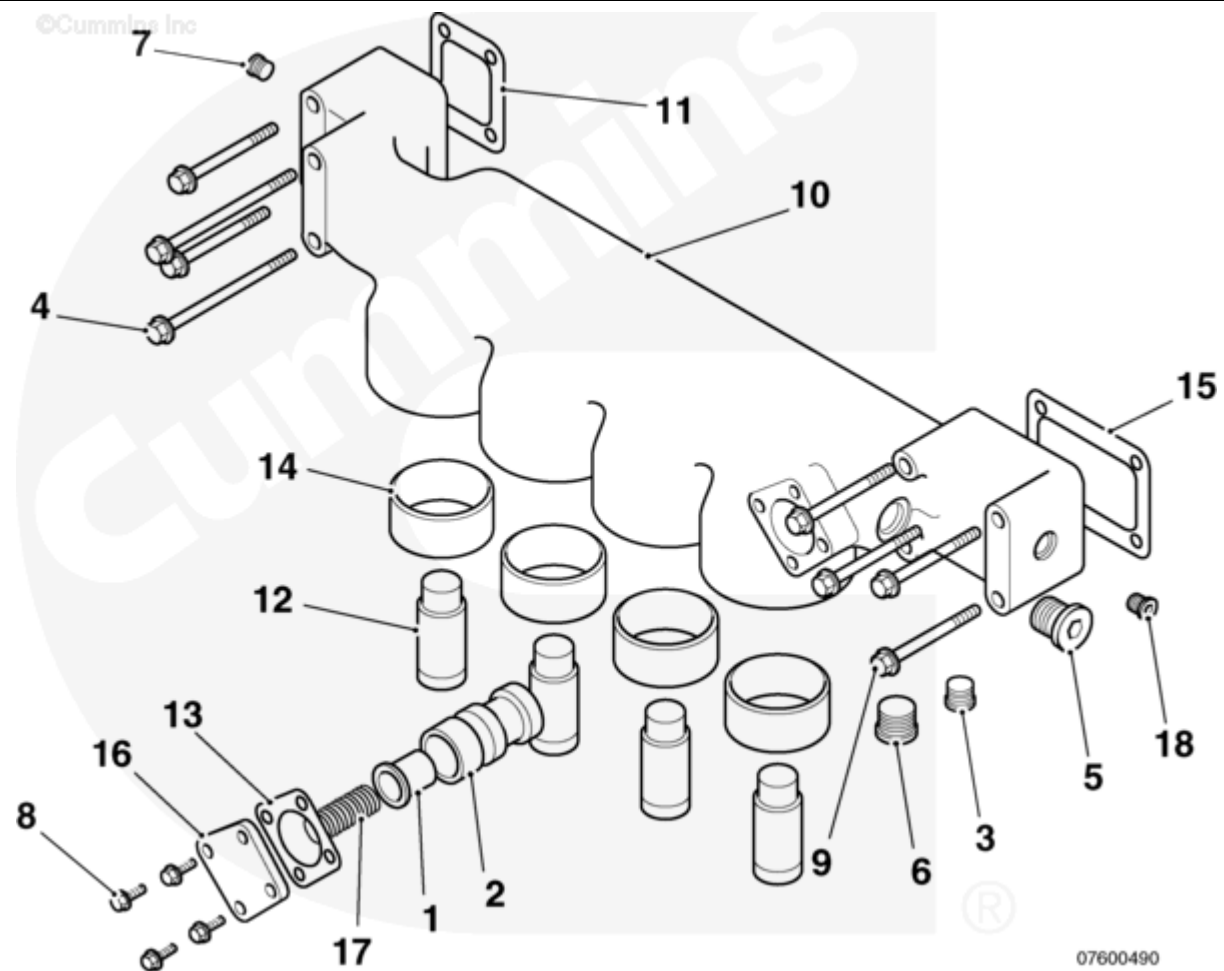
Updated One-Piece Filter Head Design

The updated one-piece filter head uses eight mounting capscrews. The previous two-piece filter head uses 10 mounting capscrews.

For the QSK60 one-piece filter head the position of the CENSE™ post oil filter pressures sensor has moved from the rear of the oil filter head facing vertically downwards to facing horizontally toward the rear of the engine.

with Electronically Actuated Injector

The fuel pump and the fuel pump drive on the QSK60 engine with electronically actuated injectors is oil lubricated.



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1. Pressure regulator plunger
2. Bypass valve body
3. Threaded plug
4. Hexagon head capscrew
5. Threaded plug
6. Threaded plug
7. Threaded plug
8. Hexagon head capscrew
9. Hexagon head capscrew
10. Lubricating oil filter head (QSK60 shown)
11. Filter head gasket
12. Filter head adapter
13. Cover plate gasket
14. Filter head adapter
15. Filter head gasket
16. Cover plate
17. Compression spring
18. Threaded plug.

